

MINUTES OF MEETING (MoM)

Clinical Validation Discussion for ICU Sentinel

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Project Reviewed: ICU Sentinel – Edge AI Based ICU Early Warning and Patient Prioritization System

The project was reviewed from a clinical perspective with emphasis on patient deterioration, early warning requirements, ICU workflow challenges, and practical deployment considerations.

1. Hemodynamic Shock Prediction

1. One of the most dangerous ICU events is progressive circulatory shock.
2. Early warning signs may include rising heart rate, narrowing pulse pressure, reduced peripheral perfusion, and gradual blood pressure decline.
3. Patients may appear clinically stable while compensating physiologically.

Example:

HR: 92 → 105 → 118 bpm

SBP: 120 → 110 → 98 mmHg

RR: 18 → 24 breaths/min

Recommendation: Predict septic shock, cardiogenic shock, and hypovolemic shock 30–60 minutes before clinical collapse whenever possible.

2. Respiratory Failure Prediction

1. Respiratory deterioration is often preceded by subtle breathing pattern changes.
2. Respiratory rate frequently rises before oxygen saturation falls significantly.
3. Delayed intervention may result in respiratory exhaustion and need for ventilatory support.

Example:

RR: 18 → 24 → 30 breaths/min

SpO₂: 98% → 95% → 92%

Recommendation: Detect patterns associated with acute respiratory failure, ARDS, pneumonia, and airway compromise 20–30 minutes before major distress.

3. Sepsis Progression Detection

1. Sepsis commonly develops gradually before becoming clinically obvious.
2. Early indicators include fever, tachycardia, increasing respiratory rate, and deterioration across multiple vital signs.
3. Earlier recognition significantly improves patient outcomes.

Example:

Temperature: 37.2°C → 38.3°C → 39.0°C

HR: 88 → 102 → 120 bpm

RR: 18 → 24 → 28 breaths/min

Recommendation: Identify patterns suggesting systemic infection and sepsis progression several

hours before organ dysfunction develops.

4. Cardiac Instability Detection

1. Cardiovascular deterioration may occur rapidly in critically ill patients.
2. Arrhythmias are often preceded by progressive physiological stress.
3. Continuous ECG analysis can provide earlier warning than conventional threshold alarms.

Recommendation: Use ECG trend analysis to support identification of worsening cardiac status and rhythm instability.

5. Multi-Parameter Trend Analysis

1. Clinical deterioration should not be assessed using a single parameter.
2. Multiple physiological variables should be evaluated together.

Example:

HR = 115 bpm

RR = 28 breaths/min

SpO₂ = 94%

Temperature = 38.7°C

A combined abnormal trend is clinically more significant than isolated tachycardia alone.

Recommendation: Analyze heart rate, respiratory rate, blood pressure, SpO₂, temperature, and ECG-derived features simultaneously.

6. ICU Prioritization Requirements

1. High patient-to-staff ratios remain a major challenge in many ICUs.
2. Important physiological changes may be missed during busy periods.
3. Patients likely to deteriorate should receive higher monitoring priority.

Recommendation: Continuously rank patients by risk score, probability of deterioration, and estimated time-to-critical-event.

7. False Alarm Management

1. Alarm fatigue is a major clinical problem.
2. Common causes of false alarms include sensor displacement, loose electrodes, motion artifacts, and signal dropouts.

Recommendation: Validate physiological plausibility and suppress obvious sensor artifacts before generating high-priority alerts.

8. Clinical Safety Considerations

1. The system should function as a decision-support tool rather than replace clinical judgment.
2. Every alert should clearly communicate the predicted deterioration type, risk level, contributing factors, and urgency.

Recommendation: Present clear, interpretable alerts that can be rapidly understood by ICU staff.

Final Clinical Recommendation

The consultant concluded that ICU sentinel addresses a clinically significant problem by shifting ICU monitoring from reactive detection to proactive prediction. Particular value was identified in early shock prediction, respiratory deterioration forecasting, sepsis progression detection, dynamic patient prioritization, and continuous Edge-AI based monitoring. Continued validation using real-world ICU datasets was recommended before deployment.